

Mersenne Scaling in Cyclic Rule 90

Route-A structural certificate C150

Abstract

For Rule 90 on every binary ring of Mersenne circumference $L = 2^r - 1$, the local multiplier $a = x + x^{-1}$ satisfies $a^{L+1} = a$. Its image contains exactly half the state space, every state reaches it in one step, and all image points are periodic with periods dividing L . Polynomial gcd and Möbius inversion resolve primitive cycles. In contrast, power-of-two rings are nilpotent. This is an exact scaling theorem, not a target determinant.

1 Mersenne identity and image

Work in $R_L = \mathbb{F}_2[x, x^{-1}]/(x^L - 1)$, where one Rule-90 update is multiplication by $a = x + x^{-1}$. If $L = 2^r - 1$, Frobenius gives

$$a^{2^r} = x^{2^r} + x^{-2^r} = x + x^{-1} = a, \quad a^{L+1} = a. \quad (1)$$

For monic f and a multiplier h , writing $f = gf_1$, $h = gh_1$ with $g = \gcd(f, h)$ shows $f \mid hq$ exactly when $f_1 \mid q$. Thus multiplication by h on $k[x]/(f)$ has kernel dimension $\deg g$. Multiplication by x is invertible and $xa = (x+1)^2$. Since odd L makes $x^L + 1$ squarefree and $x+1$ is a factor, $\gcd(x^L + 1, (x+1)^2) = x+1$. Hence $\dim \ker a = 1$ and $\dim \operatorname{im} a = L - 1$.

For $y = au$ in the image, (1) yields $a^L y = y$. Thus the restriction to the image is a permutation with order dividing L . Every state enters the image after one update. Conversely, if $a^n u = u$, then $u = a(a^{n-1}u)$ lies in the image. The periodic set is therefore exactly the image, with 2^{L-1} points: exactly half of all states.

2 Exact periods

The standard multiplication-kernel calculation gives

$$\operatorname{Fix}_L(n) = 2^{\deg \gcd(x^L + 1, (x^2 + 1)^n + x^n)}. \quad (2)$$

Consequently

$$P_L(n) = \sum_{d \mid n} \mu(n/d) \operatorname{Fix}_L(d), \quad C_L(n) = P_L(n)/n.$$

All realized periods divide L . The converse is not asserted: a divisor of L need not be realized. The division by n occurs only after Möbius inversion, so fixed configurations, exact-period configurations, and primitive cycles remain distinct.

3 Power-of-two control and boundary

For $L = 2^s$, $a^{2^{s-1}} = x^{2^{s-1}} + x^{-2^{s-1}} = 0$ because the two monomials coincide. Thus the map is nilpotent and only zero is periodic. Indeed, if $a^n u = u$, repeated use of this equality expresses u as $a^{qn}u$ for every q ; choosing q beyond the nilpotency index forces $u = 0$.

The structural statements above hold for every $r, s \geq 1$; the exact ledgers through $r, s \leq 8$ are implementation sentinels. They contain 27 divisor-period cells and pass 153 independent matrix assertions, 276 SymPy checks, byte replay, and 45 hostile cases. The strict tuple is (A1_WEAK, A2_FAIL, A3_FAIL, A4_FAIL). We claim no infinite-volume determinant or thermodynamic limit, target divisor, target functional equation or counting law, arithmetic/local factor, root number, automorphy, natural operator lift, Hilbert–Pólya operator, or Route-B authorization. Scope: NO_BAD_EULER_OR_ROOT_NUMBER.